AgriBuddy – Smart Crop Scheduling & Advisory Web Application

Project Report | Software Engineering Lab

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Problem Statement ID: 25010

1. PROJECT TITLE

AgriBuddy – Smart Crop Scheduling & Advisory Web Application

2. PROBLEM STATEMENT

Small and marginal farmers in India often lack access to structured, data-driven agricultural planning tools. Most farming decisions — including irrigation scheduling, fertilizer application, and crop planning — are made based on personal experience or irregular local advice, rather than real-time weather data and scientifically validated soil information.

The key challenges faced by farmers today include:

Unpredictable Climate Impact: Erratic rainfall patterns, unexpected temperature changes, and unseasonal weather events severely affect crop yield. Without real-time weather integration, farmers cannot plan irrigation or fertilization activities effectively.

Lack of Digital Record-Keeping: Farmers do not maintain systematic logs of their farming activities such as sowing dates, fertilizer usage, irrigation events, or pest treatments. This makes it impossible to analyze patterns, optimize inputs, or improve productivity over seasons.

Absence of Personalized Advisory: Generic agricultural advice does not account for a farmer's specific soil type, crop variety, local climate, or farm size. Without personalized guidance, farmers often over-irrigate, over-fertilize, or miss critical crop care windows.

Inefficient Resource Usage: Due to the absence of smart scheduling, farmers waste significant amounts of water and fertilizer, increasing input costs and degrading soil health over time.

Limited Access to Technology: Rural and semi-urban farmers typically have access only to basic smartphones or low-bandwidth internet. Existing digital agricultural tools are either too complex or unavailable in regional languages.

There is therefore a strong need for a smart, accessible, and localized advisory platform that integrates weather data, soil information, and crop schedules to support informed, data-driven agricultural decision-making — particularly for small and marginal farmers across India.

3. OBJECTIVES

The AgriBuddy platform is built to fulfill the following primary and secondary objectives:

Primary Objectives:

1. To develop a responsive web application that assists small and marginal farmers with intelligent crop scheduling and agricultural decision-making.
2. To integrate real-time weather data using APIs such as OpenWeatherMap and generate automated irrigation and fertilization alerts based on forecast conditions.
3. To build a smart crop calendar that provides farmers with a personalized schedule of planting stages, recommended practices, and resource inputs for each crop.
4. To create a digital crop diary that allows farmers to log and track daily farming activities including sowing, watering, fertilizer application, and harvesting.
5. To implement a crop advisory system that delivers localized, data-driven recommendations based on crop type, soil condition, seasonal patterns, and weather forecasts.
6. To develop a notification system that sends timely in-app, email, or SMS alerts to farmers regarding critical agricultural events and advisories.

7. To design a multilingual, mobile-friendly user interface that is accessible to farmers in rural and semi-urban areas with minimal technical knowledge.
8. To provide an analytics dashboard with visual insights into crop timelines, weather summaries, input usage, and yield predictions.
9. To maintain a structured and normalized relational database that stores user profiles, farm data, crop schedules, activity logs, weather records, and advisories.
10. To support two user roles — Farmer and Agricultural Advisor/Admin — with appropriate access controls and privileges.

Expected Outcomes:

- Improved crop planning and resource optimization across seasons.
- Reduced water and fertilizer wastage through smart scheduling.
- Increased crop productivity through timely and personalized advisory.
- Establishment of a digital record management system for farms.
- Better agricultural decision-making through real-time data integration.

4. METHODOLOGY / APPROACH

4.1 Development Methodology: Agile Scrum Framework

AgriBuddy adopts the **Scrum (Agile)** methodology as its software development approach. This choice is justified by the nature of the project — AgriBuddy consists of multiple interdependent modules such as user management, weather API integration, crop advisory logic, notification system, and analytics dashboard. These modules can be developed, tested, and validated incrementally, making iterative development ideal.

Agile ensures:

- Incremental delivery of working software at the end of each sprint.
- Continuous validation of features against requirements.
- Flexibility to refine advisory logic and UI based on testing feedback.
- Structured milestone monitoring throughout the 10-week project timeline.

4.2 Scrum Roles

Product Owner: Defines system requirements, prioritizes the product backlog, and validates sprint deliverables to ensure alignment with project objectives.

Scrum Master: Facilitates sprint planning, monitors progress, ensures adherence to Agile principles, and resolves blockers during development.

Development Team: Responsible for frontend development, backend API development, database design, API integration, advisory logic implementation, testing, and deployment. In this academic context, these roles are logically separated but executed by the same individual.

4.3 Product Backlog

The prioritized product backlog for AgriBuddy includes: User Registration and Login, Farmer Profile Management, Crop Diary Module, Weather API Integration, Soil Data Handling, Crop Scheduling Logic, Advisory and Alert System, Notification System, Analytics Dashboard, Admin Panel, and Testing & Deployment. Backlog refinement is performed before each sprint to ensure proper prioritization.

4.4 Sprint Plan & Milestones

Sprint 1 (Weeks 1–2) — Core Setup & User Management

Goal: Establish the foundation of the system. Tasks: Finalize requirements, set up frontend and backend project structure, design the relational database schema, implement user registration and login, and create the farmer profile module. Deliverable: Functional authentication system with profile management. Milestone 1: User management module implemented and tested.

Sprint 2 (Weeks 3–4) — Data Integration

Goal: Enable real-time data flow into the system. Tasks: Integrate Weather API (OpenWeatherMap), create soil data input functionality, design crop entity and database

relationships, and connect frontend with backend APIs. Deliverable: System successfully fetches and displays real-time weather data. Milestone 2: Weather API integration and data flow validation completed.

Sprint 3 (Weeks 5–7) — Advisory Logic & Notifications

Goal: Build the core intelligence of the system. Tasks: Implement the crop scheduling engine, develop irrigation and fertilizer advisory rules, implement crop diary logging, and add the notification system (in-app/email). Deliverable: Automated advisory system based on weather and crop data. Milestone 3: Advisory and alert system fully functional.

Sprint 4 (Weeks 8–10) — Testing, Dashboard & Deployment

Goal: Complete, test, and deploy the full system. Tasks: Develop the analytics dashboard, implement the admin panel, perform unit and integration testing, fix identified bugs, deploy the application, and prepare documentation. Deliverable: Complete working AgriBuddy web application. Milestone 4: System deployed and ready for demonstration.

4.5 Incremental Product Versions

Version	Output
v0.1	User authentication module
v0.2	Weather integration module
v0.3	Advisory logic module
v1.0	Fully functional deployed system

4.6 Technology Stack

Layer	Technology
Frontend	HTML, CSS, JavaScript (React / Vue.js)
Backend	Python (Flask / Django) or Node.js
Database	MySQL / PostgreSQL / SQLite
External API	OpenWeatherMap API
Notifications	SMTP (Email) / SMS Gateway
Deployment	Heroku / Render / Firebase
Architecture	RESTful Client-Server System

4.7 System Architecture

The AgriBuddy architecture follows a layered client-server model comprising the following components:

User Interface Modules: The frontend layer accessed by farmers and advisors via mobile or desktop browsers. It includes the Dashboard, Crop Input Forms, Scheduling Views, and Advisory Display Pages.

Backend Server: Handles all business logic, API routing, authentication, session management, data validation, and advisory generation. Built using Flask/Django or Node.js.

Crop Advisory Logic Engine: The core intelligence module that processes weather data, soil conditions, and crop schedules to generate actionable recommendations and alerts.

Database Layer: A normalized relational database (MySQL/PostgreSQL) storing all entities including Users, Farms, Crops, CropSchedules, ActivityLogs, WeatherData, Advisories, and Notifications.

Weather API Integration: Real-time weather data is fetched from OpenWeatherMap and used to trigger irrigation/fertilization alerts and advisory generation.

Notification System: Sends event-triggered and scheduled alerts through in-app notifications, email (SMTP), or SMS gateways.

4.8 Database Design

The system uses a fully normalized relational database adhering to Third Normal Form (3NF). The key entities and their relationships are:

User: Stores user_id, user_type (farmer/advisor), user_name, email, password, phone_number, address, and language_preference. One user can own multiple farms.

Farm: Stores farm_id, farm_size, location, soil_type, and date_created. Each farm belongs to one user and can have multiple crop schedules and weather records.

Crop: Stores crop_id, crop_name, ideal_soil_type, crop_duration, season, water_requirement, crop_type, and date_created.

CropSchedule: Links a farm to a crop with planting_date, expected harvest date, yield estimate, and status. Each schedule can generate multiple activity logs.

CropActivityLog: Records daily activities (irrigation, fertilization, pest treatment) linked to a specific crop schedule.

WeatherData: Stores location-based weather readings including temperature, humidity, rainfall prediction, wind speed, and weather condition.

Advisory: Expert recommendations linked to crops and users, containing advisory_type, content, priority, and language.

Notification: System-generated alerts sent to users, linked to advisories with status tracking.

4.9 Agile Ceremonies

Each sprint in AgriBuddy includes the following ceremonies: Sprint Planning (define sprint goals and assign tasks), Daily Stand-up (monitor progress and resolve blockers), Sprint Review (demonstrate completed functionality), and Sprint Retrospective (identify improvements for the next sprint).

4.10 Risk Management

Risk	Mitigation Strategy
Weather API failure	Implement error handling and retry mechanisms
Development delays	Reprioritize backlog tasks in next sprint
Integration errors	Perform end-to-end testing after each sprint
Low-bandwidth access	Implement offline caching and SMS-based alerts
Data security breach	Enforce HTTPS, encryption, and RBAC

5. WORK COMPLETED TILL NOW

The AgriBuddy project has achieved significant progress across all planned phases of development. As of the current sprint, approximately **50% of the total project implementation has been completed**, covering the full planning, design, documentation, and frontend development phases. Below is a detailed account of all work completed so far.

5.1 Week 0 — Problem Identification & Project Ideation

The foundation of the project was established by identifying the core agricultural problem faced by small and marginal farmers in India. The problem statement was formally documented with Problem Statement ID 25010, clearly describing the challenges of lack of real-time data, absence of personalized advisory, and poor digital record-keeping. The proposed solution — AgriBuddy — was defined with its key features, expected outcomes, and stakeholder mapping covering farmers, agricultural experts, government departments, and rural development organizations.

5.2 Week 1 — Architecture Design & Software Requirements Specification

The system architecture was designed showing the complete data flow between the User Interface, Backend Server, Crop Advisory Logic Engine, Weather API, and the Database layer. The full Software Requirements Specification (SRS) document was completed, covering product purpose, scope, user classes (Farmer and Agricultural Expert/Admin), functional requirements, external interface requirements, and non-functional requirements including performance, security, safety, and software quality attributes. The Entity-Relationship (ER) Diagram was created defining all eight core entities and their relationships. The distributed database model was designed to ensure high availability across rural regions. Both a Deliverable-Based and Phase-Based Work Breakdown Structure (WBS) were produced, along with a comprehensive Gantt Chart mapping all development activities across the 10-week timeline.

5.3 Week 2 — Agile Process Methodology Documentation

The complete Agile Scrum methodology framework was documented for the project. Scrum roles were defined, the full product backlog with 11 prioritized features was created, and all 4 sprints were planned with specific goals, tasks, deliverables, and milestones. Incremental version planning (v0.1 to v1.0) was established and risk management strategies within the Agile framework were outlined. This week's output formally established the development process to be followed throughout the project lifecycle.

5.4 Week 3 — UML Design: Use Case & Class Diagrams

The Use Case Diagram was developed modeling two actors — Farmer and Advisor — and nine use cases: RegisterOrLogin, ManageProfile, ManageFarms, AddOrEditCropSchedule, LogCropActivity, ViewWeatherData, GenerateAdvisories, ReceiveNotifications, and ViewAdvisories. Include and extend relationships were precisely defined. The Class Diagram was completed modeling all eight system entities (User, Farm, Crop, CropSchedule, CropActivityLog, WeatherData, Advisory, Notification) with full attributes, data types, visibility modifiers, and multiplicity-based associations. This formed the complete structural blueprint of the system's data architecture.

5.5 Week 4 — UML Behavioral Diagrams: Sequence & Activity Diagrams

The Sequence Diagram was created for the "Log Crop Activity" use case, modeling 10 sequential interactions across five system participants: User, User Interface, System Controller, Crop Activity Logs, and Notification System — covering user validation, data persistence, confirmation messaging, and notification triggering. The Activity Diagram for the complete system workflow was developed, mapping all user actions from Login/Register through Dashboard navigation (Manage Farms, Add/Edit Crop Schedule, Log Crop Activity, View Advisories) to Logout, including all conditional branches, validation checks, and error-retry flows.

5.6 Week 5 — Frontend Development (Currently In Progress — 50% Implementation Milestone)

The frontend of AgriBuddy has been fully developed as a responsive web application. The following pages and modules have been implemented and are fully functional on the frontend:

Landing Page: A professional home page showcasing AgriBuddy's value proposition with a hero section, live crop dashboard preview widget, platform statistics (15,000+ farmers, 75,000+ crops managed, 98% accuracy rate, 24/7 weather monitoring), and a feature highlights section covering Real-time Weather, Smart Crop Calendar, Crop Diary, Localized Recommendations, Analytics & Reports, and Multi-language Support.

User Authentication — Login Page: A role-based login system allowing users to sign in as Farmer or Agricultural Expert, with email and password fields, secure input handling, and navigation to registration.

User Registration Page: A complete account creation form capturing Name, Email, Password, Phone Number, Role (Farmer/Expert), Location, and Farming Experience — establishing the user profile data model in the frontend.

Farmer Dashboard: A fully functional main dashboard for logged-in farmers displaying navigation modules (My Farms, Crop Diary, Weather, Analytics, Ask Expert, AI Assistant), farm statistics (Total Farms, Total Area, Active Crops, Health Score), live current weather widget (28°C, Partly Cloudy, Humidity 65%, Wind Speed 12 km/h), and a Yield Trend graph plotting monthly crop yield data from January to June.

My Farms Page: A farm management interface displaying registered farm cards with details including location, area in acres, soil type, irrigation method, active crops and their growth status. Supports adding new farms via the "+ Add New Farm" button.

Weather Module: A dedicated weather page displaying current conditions (temperature, humidity, wind speed, pressure, visibility), weather alerts with advisory messages (e.g., "Rain Expected — consider covering sensitive crops"), and a 5-day weather forecast with rain probability percentages for each day.

Analytics Dashboard: A comprehensive farm performance insights page displaying total revenue (₹1,25,000), total profit (₹40,000), total yield (2400 kg), and profit margin (32%) with trend indicators. Includes a Yield-Revenue Trend area chart (Jan–Jun), Crop Distribution pie chart (Rice 45%, Wheat 30%, Corn 15%, Vegetables 10%), Cost Breakdown panel (Seeds, Fertilizer, Water, Labor, Equipment, Other), Efficiency Metrics progress bars (Water Usage, Fertilizer Efficiency, Yield per Acre, Cost Efficiency), and a Recommendations section.

Expert Connect — Farmer Side: A query management interface where farmers can view, search, and post queries to agricultural experts. Displays query status (open/in progress), urgency level, and query descriptions. Includes an "Ask Expert" button for posting new queries.

Agricultural Expert — Welcome Dashboard: A dedicated landing view for expert users showing open queries count, AI Assistant availability, and notification status. Displays recent farmer queries with farmer name, location, date, and a "Respond" action button.

Agricultural Expert — Full Expert Dashboard: A detailed consultation management interface showing total consultations (15), in-progress cases (3), resolved cases (10), average rating (4.5), and average response time (3h). Features tabbed navigation between Open Queries, My Consultations, and Resolved cases, with search and filter options. Includes a side panel showing full consultation detail — farmer name, location, query description, conversation thread, response input box, and a "Mark as Resolved" button.

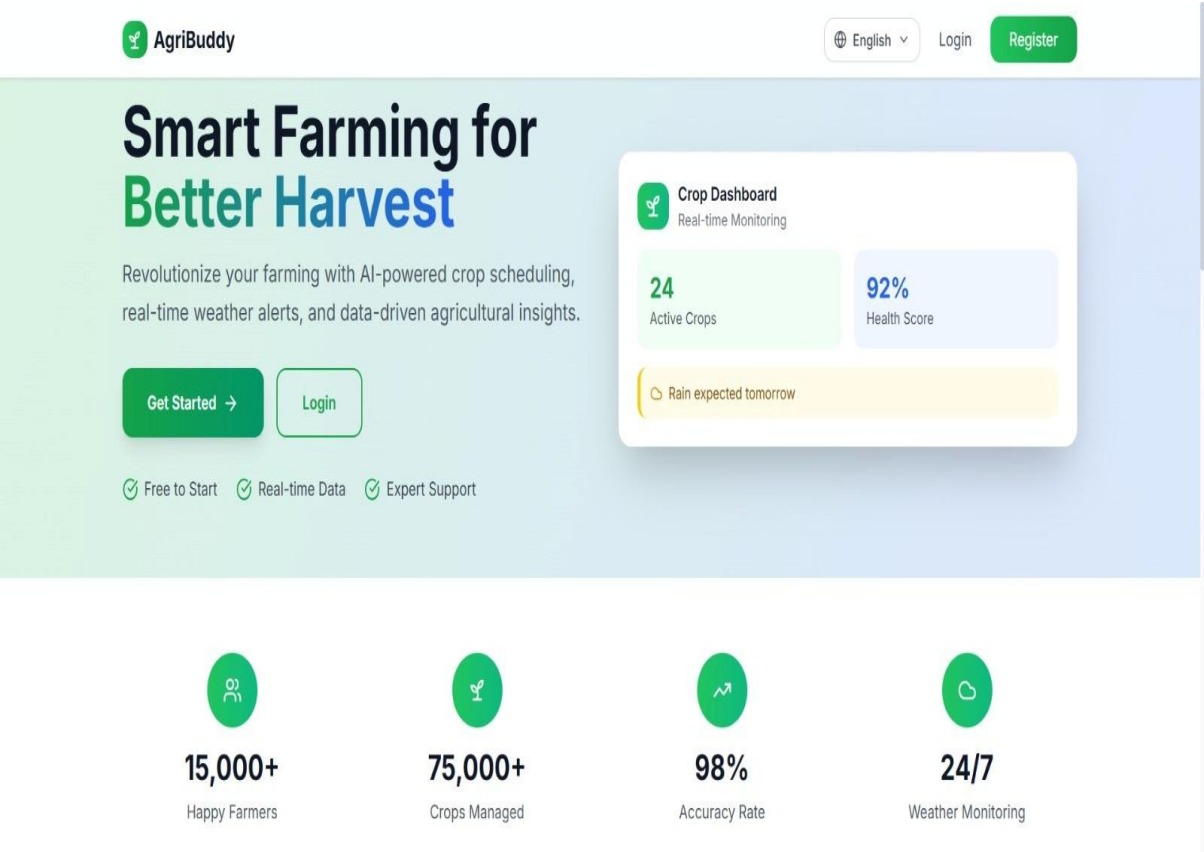
5.7 Summary of Completion Status

Phase	Status
Problem Statement & Ideation	Completed
SRS & Architecture Design	Completed
Agile Methodology Documentation	Completed
Use Case & Class Diagrams	Completed
Sequence & Activity Diagrams	Completed
Frontend Development (All Pages)	Completed

Phase	Status
Backend API Development	In Progress (Sprint 3)
Database Integration	In Progress (Sprint 3)
Advisory Logic Engine	Planned (Sprint 3)
Testing & Deployment	Planned (Sprint 4)

6 .Screenshots

6.1 Landing Page — Home Screen



Description:

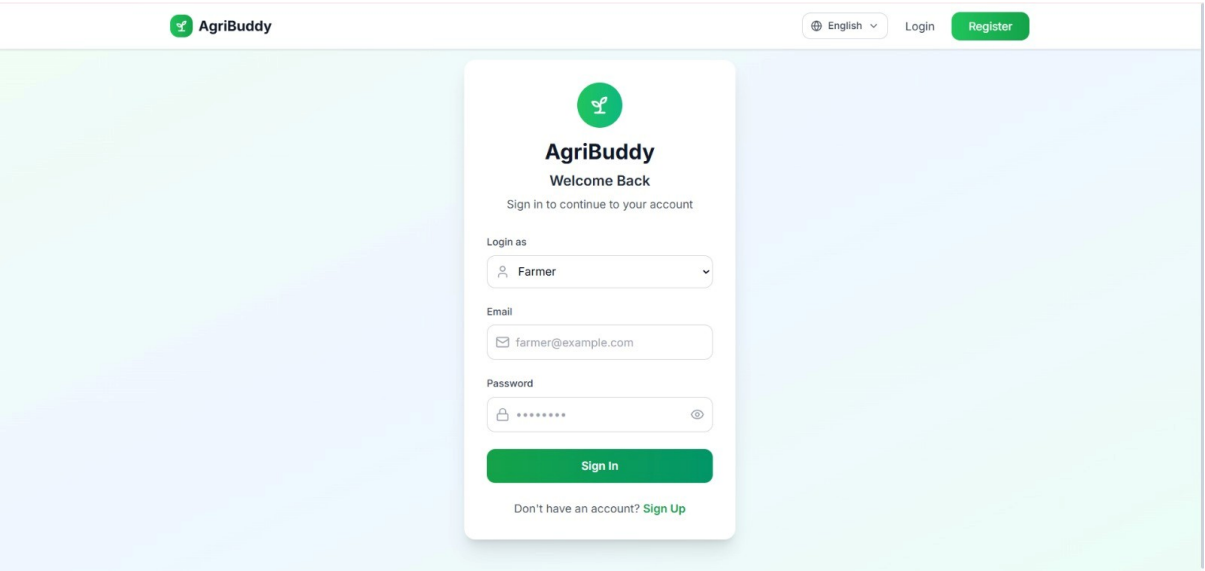
The AgriBuddy landing page is the public-facing entry point of the application. The hero section displays the tagline "Smart Farming for Better Harvest" along with a live Crop Dashboard preview widget showing 24 Active Crops and a 92% Health Score with a rain alert. Below the hero, four platform statistics are highlighted — 15,000+ Happy Farmers, 75,000+ Crops Managed, 98% Accuracy Rate, and 24/7 Weather Monitoring — reinforcing the platform's credibility and reach to new users.

6.2 Features Section — Why Choose AgriBuddy?

Description:

This section of the landing page presents the six core features of AgriBuddy in a clean card-based grid layout. The features highlighted are Real-time Weather (accurate forecasts and alerts for crop planning), Smart Crop Calendar (personalized crop stage tracking with resource inputs), Crop Diary (digital logbook from sowing to harvesting), Localized Recommendations (crop suggestions based on location and soil type), Analytics & Reports (visual insights for yield trends and input efficiency), and Multi-language Support (available in 9 Indian languages). Each feature card uses a distinct green icon, making the interface visually intuitive and accessible for rural farmers.

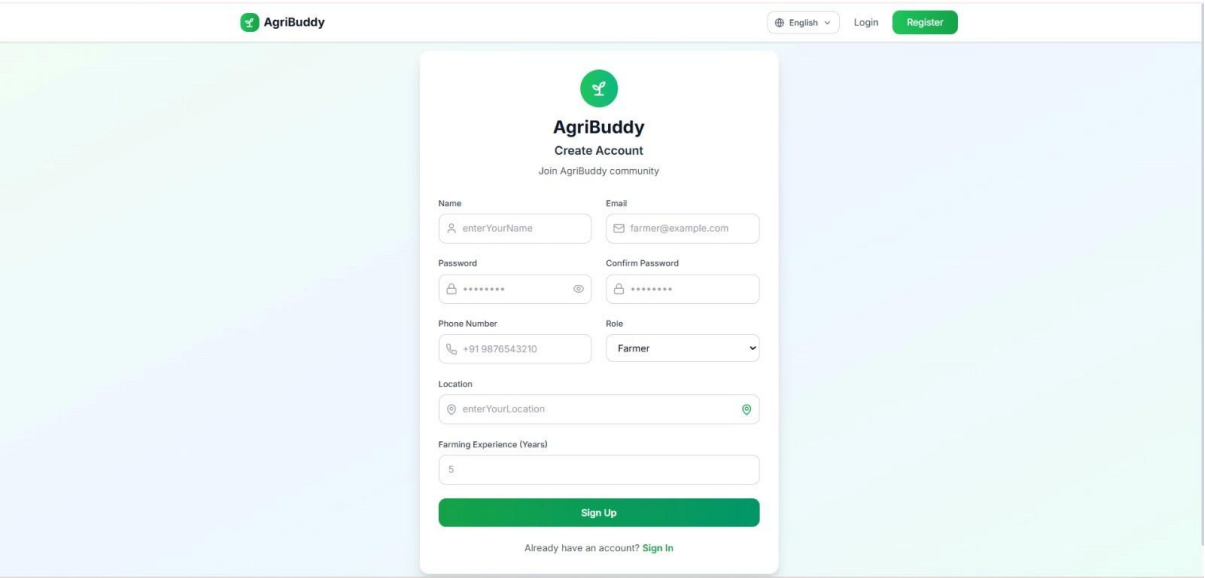
6.3 Login Page — User Authentication



Description:

The Login page provides secure role-based authentication for AgriBuddy users. It features a "Login as" dropdown allowing users to select their role (Farmer or Agricultural Expert), along with Email and Password input fields with a password visibility toggle. A green "Sign In" button submits the credentials, and new users are redirected to Sign Up via the link at the bottom. The clean centered card layout on a soft gradient background maintains AgriBuddy's consistent brand identity throughout the application.

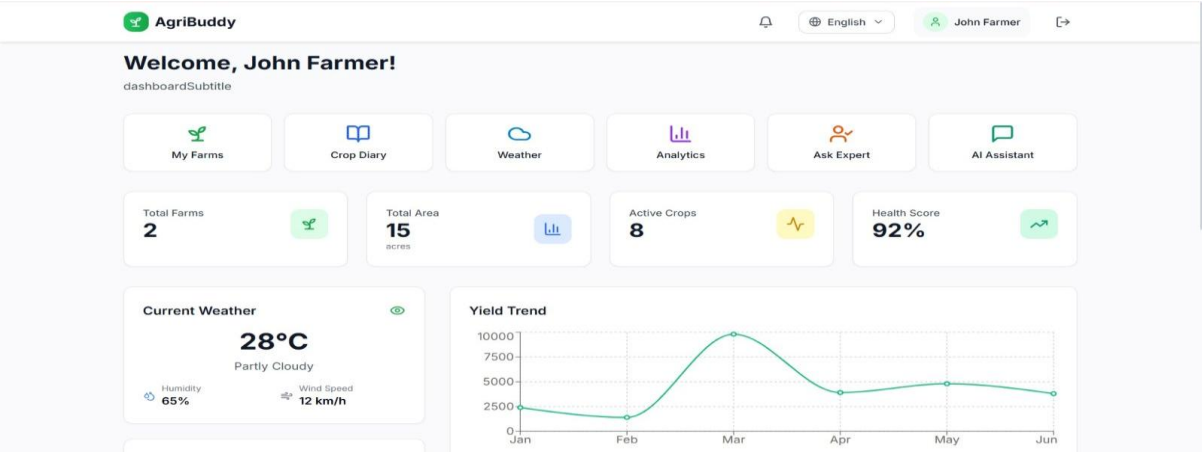
6.4 Registration Page — Create Account



Description:

The Registration page allows new users to create their AgriBuddy account by filling in details such as Name, Email, Password, Confirm Password, Phone Number, Role (Farmer/Expert), Location, and Farming Experience in years. This form captures all essential user profile data required to personalize the farming advisory experience. A "Sign Up" button submits the form, and existing users are redirected to Sign In via the link at the bottom. The two-column form layout ensures a clean and organized data entry experience consistent with the overall application design.

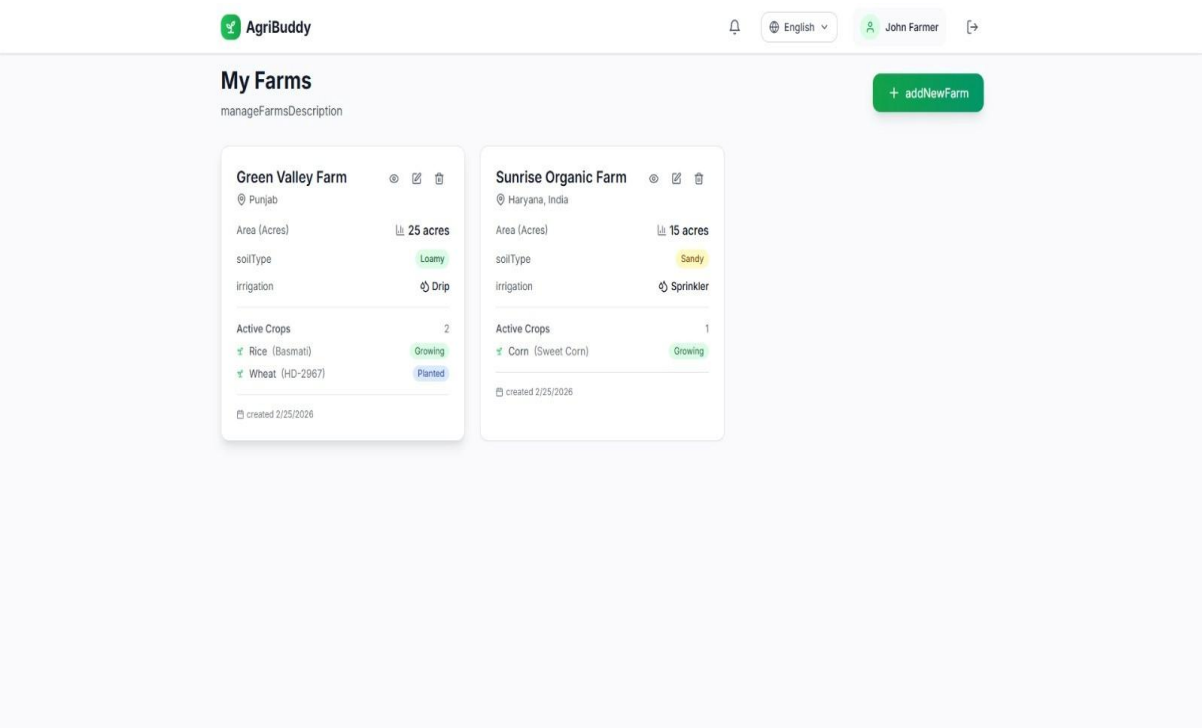
6.5 Farmer Dashboard — Main Home Screen



Description:

The Farmer Dashboard serves as the central hub for logged-in farmers, providing a complete overview of all farm activities at a glance. Six navigation modules are displayed at the top — My Farms, Crop Diary, Weather, Analytics, Ask Expert, and AI Assistant — giving quick access to all system features. Key farm statistics show Total Farms (2), Total Area (15 acres), Active Crops (8), and Health Score (92%), while a live Current Weather widget displays 28°C Partly Cloudy with Humidity 65% and Wind Speed 12 km/h. A Yield Trend line graph plots monthly crop yield from January to June, peaking at approximately 10,000 units in March, helping farmers visually track their productivity over time.

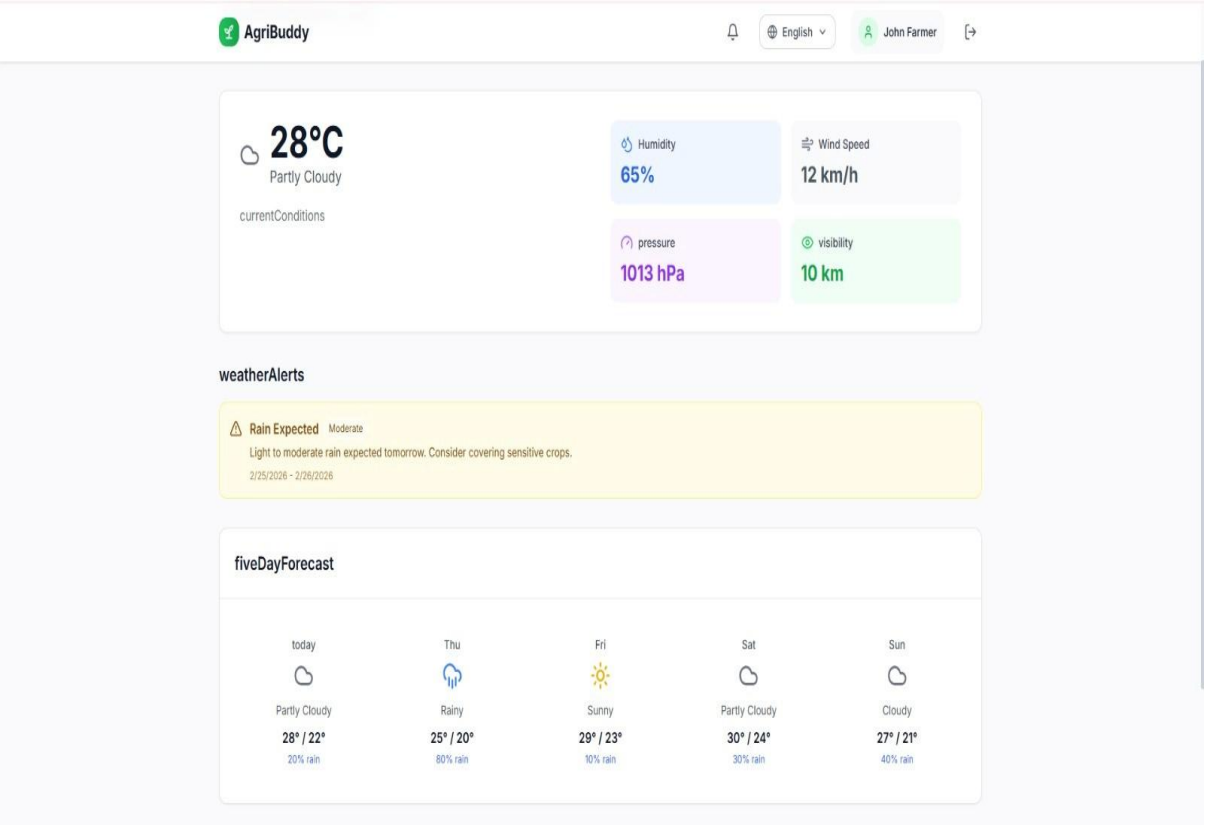
6.6 My Farms Page — Farm Management



Description:

The My Farms page displays all registered farms belonging to the logged-in farmer in a card-based layout. Two farm cards are shown — "Green Valley Farm" (Punjab, 25 acres, Loamy soil, Drip irrigation) with active crops Rice Basmati (Growing) and Wheat HD-2967 (Planted), and "Sunrise Organic Farm" (Haryana, 15 acres, Sandy soil, Sprinkler irrigation) with Sweet Corn currently Growing. Each card includes view, edit, and delete action icons, and a green "+ Add New Farm" button at the top right allows farmers to register additional farms to the system.

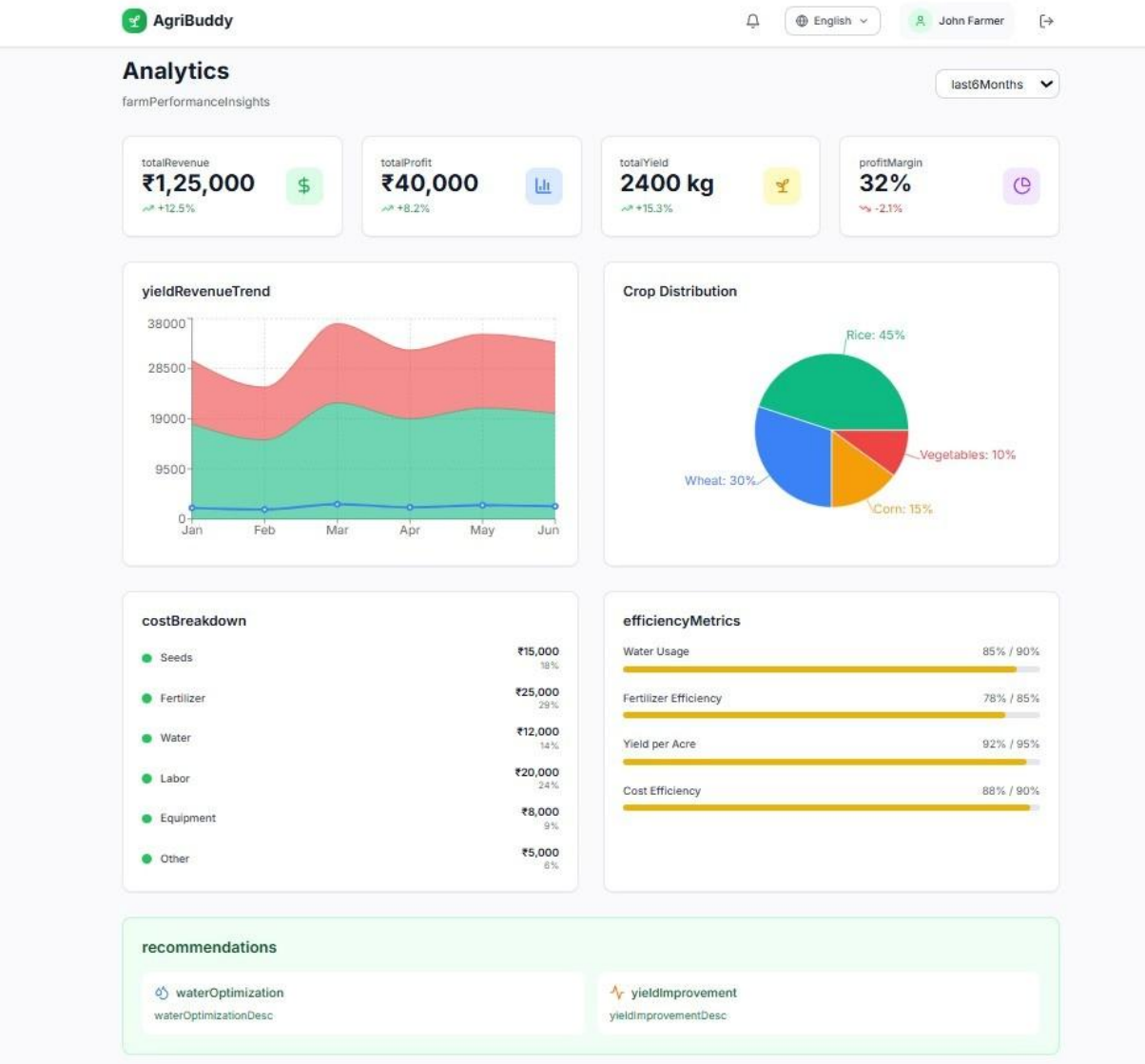
6.7 Weather Module — Real-Time Weather & Forecast



Description:

The Weather page provides farmers with comprehensive real-time meteorological data to support informed crop management decisions. The current conditions panel displays 28°C Partly Cloudy along with Humidity (65%), Wind Speed (12 km/h), Pressure (1013 hPa), and Visibility (10 km). A Weather Alerts section highlights a moderate rain warning — "Light to moderate rain expected tomorrow, consider covering sensitive crops" — directly linking weather events to actionable farming advice. The 5-Day Forecast section below shows daily weather icons, high/low temperatures, and rain probability for each day from today through Sunday, enabling farmers to plan irrigation and field activities in advance.

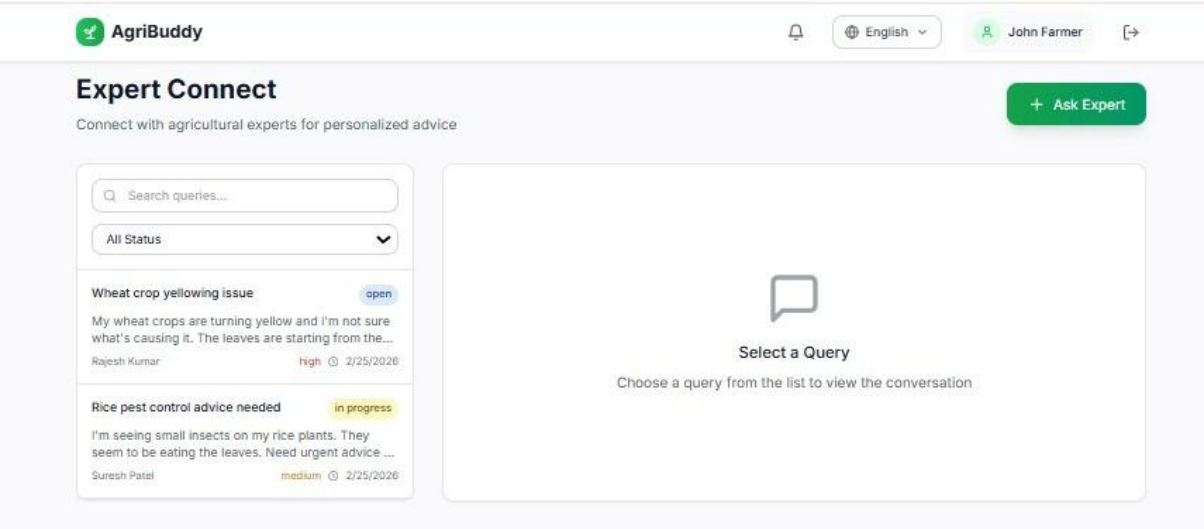
6.8 Analytics Dashboard — Farm Performance Insights



Description:

The Analytics Dashboard provides a comprehensive financial and operational overview of the farmer's performance over the last 6 months. The top summary cards display Total Revenue (₹1,25,000 with +12.5% growth), Total Profit (₹40,000 with +8.2% growth), Total Yield (2400 kg with +15.3% growth), and Profit Margin (32%). A Yield-Revenue Trend area chart tracks monthly performance from January to June, while a Crop Distribution pie chart shows Rice (45%), Wheat (30%), Corn (15%), and Vegetables (10%). The Cost Breakdown panel categorizes all input expenses across Seeds, Fertilizer, Water, Labor, Equipment, and Other, and the Efficiency Metrics section displays progress bars for Water Usage, Fertilizer Efficiency, Yield per Acre, and Cost Efficiency — each compared against their target values to highlight improvement areas.

6.9 Expert Connect Module



Description:

The Expert Connect feature allows farmers to directly connect with agricultural experts for personalized crop advice. Users can raise queries related to crop diseases, pest control, or farming issues and track their status (Open/In Progress). Each query shows priority level, submission date, and farmer name, enabling organized and timely expert responses through a clean chat-based interface.

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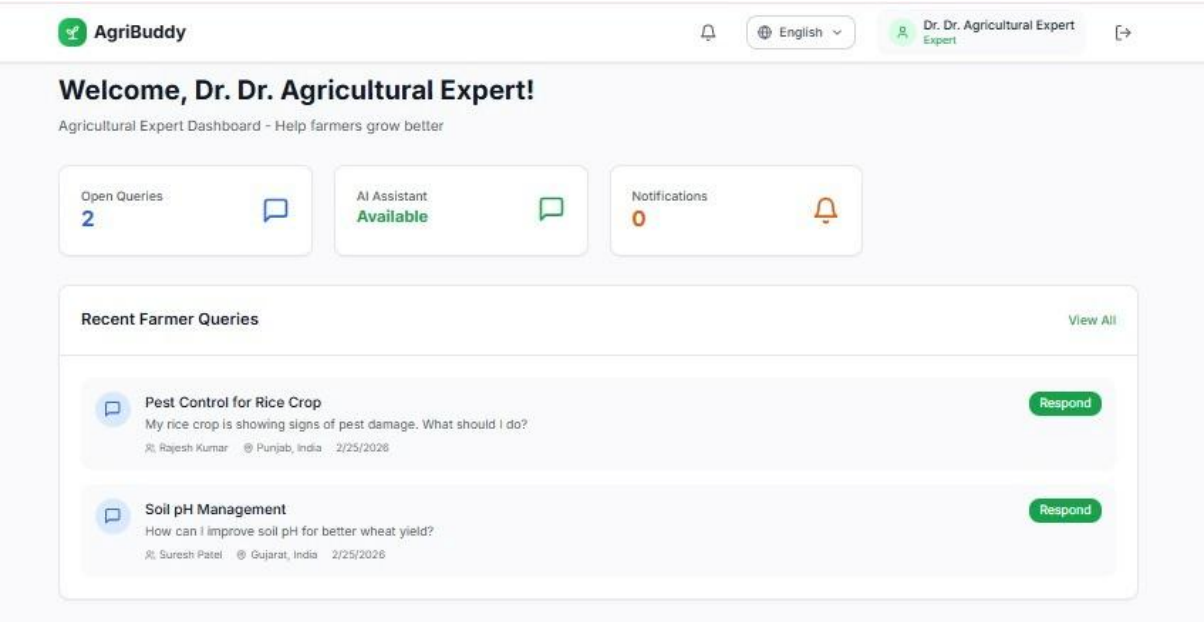
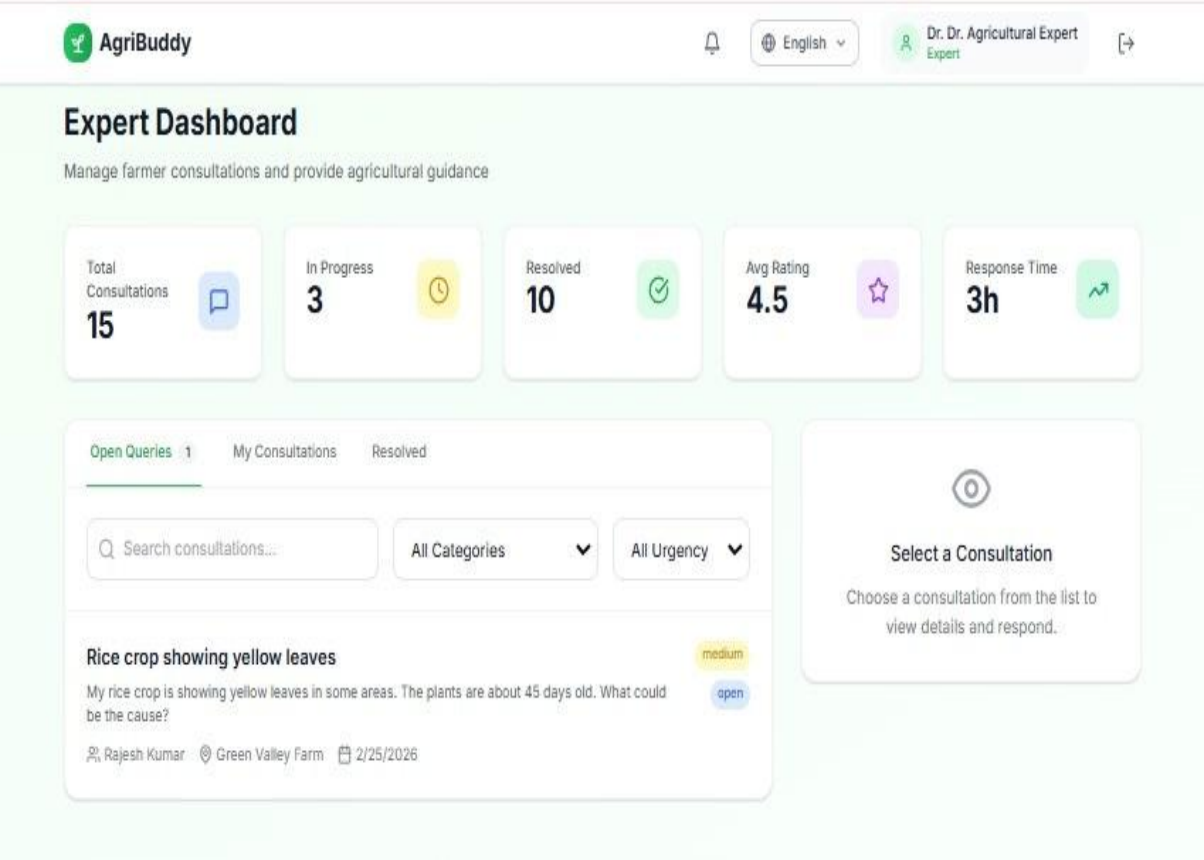
6.10 Expert Dashboard — Agricultural Expert Home Panel



Description:

The Expert Dashboard welcomes the logged-in agricultural expert and provides a quick overview of pending farmer queries and system status. Summary cards display 2 Open Queries, AI Assistant availability, and 0 Notifications. The "Recent Farmer Queries" section lists two active queries — "Pest Control for Rice Crop" by Rajesh Kumar (Punjab) and "Soil pH Management" by Suresh Patel (Gujarat), both dated 25/02/2026 — each with a green "Respond" button allowing experts to directly reply and assist farmers in real time.

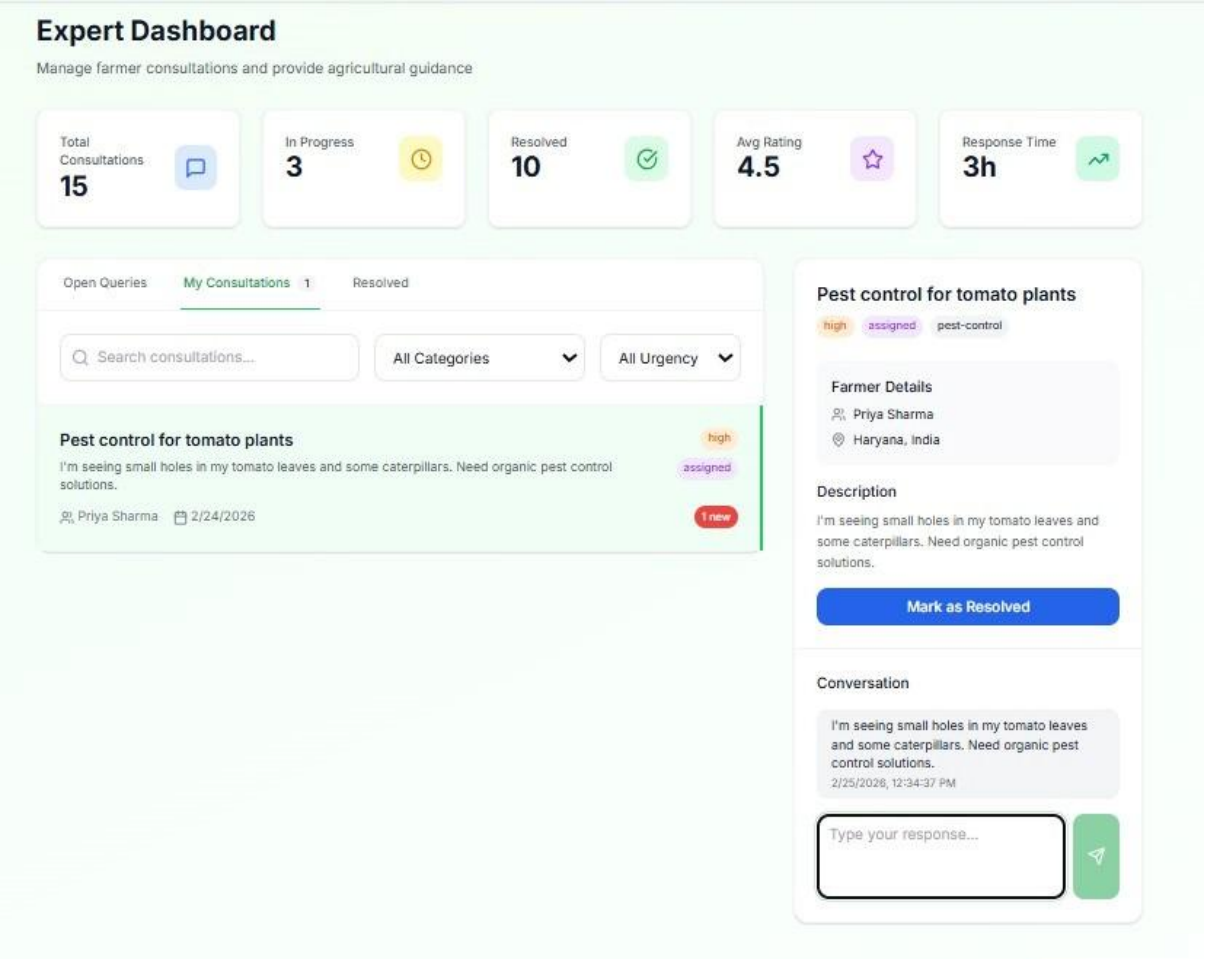
6.11 Expert Dashboard — Consultation Management Panel



Description:

The Expert Dashboard provides a comprehensive overview of all farmer consultations handled by the agricultural expert. Five summary cards display key metrics — 15 Total Consultations, 3 In Progress, 10 Resolved, 4.5 Avg Rating, and 3h Response Time. The tabbed query section (Open Queries, My Consultations, Resolved) allows easy navigation, with an active open query — "Rice Crop Showing Yellow Leaves" by Rajesh Kumar from Green Valley Farm (Medium priority, Open) — visible for the expert to review and respond to.

6.12 Expert Dashboard — My Consultations & Query Response Panel



Description:

The "My Consultations" tab displays queries assigned to the expert, with an active case — "Pest Control for Tomato Plants" (High priority, Assigned) submitted by Priya Sharma from Haryana on 24/02/2026. The right panel shows full farmer details, query description, and a live conversation thread where the expert can type and send responses directly. A "Mark as Resolved" button allows the expert to close the consultation once the issue is addressed, enabling smooth end-to-end query resolution.

Demo Video

https://drive.google.com/file/d/1D1rsRPIGKUpBfLNXjGVnRHWqJNn-fPg1/view?usp=sharing